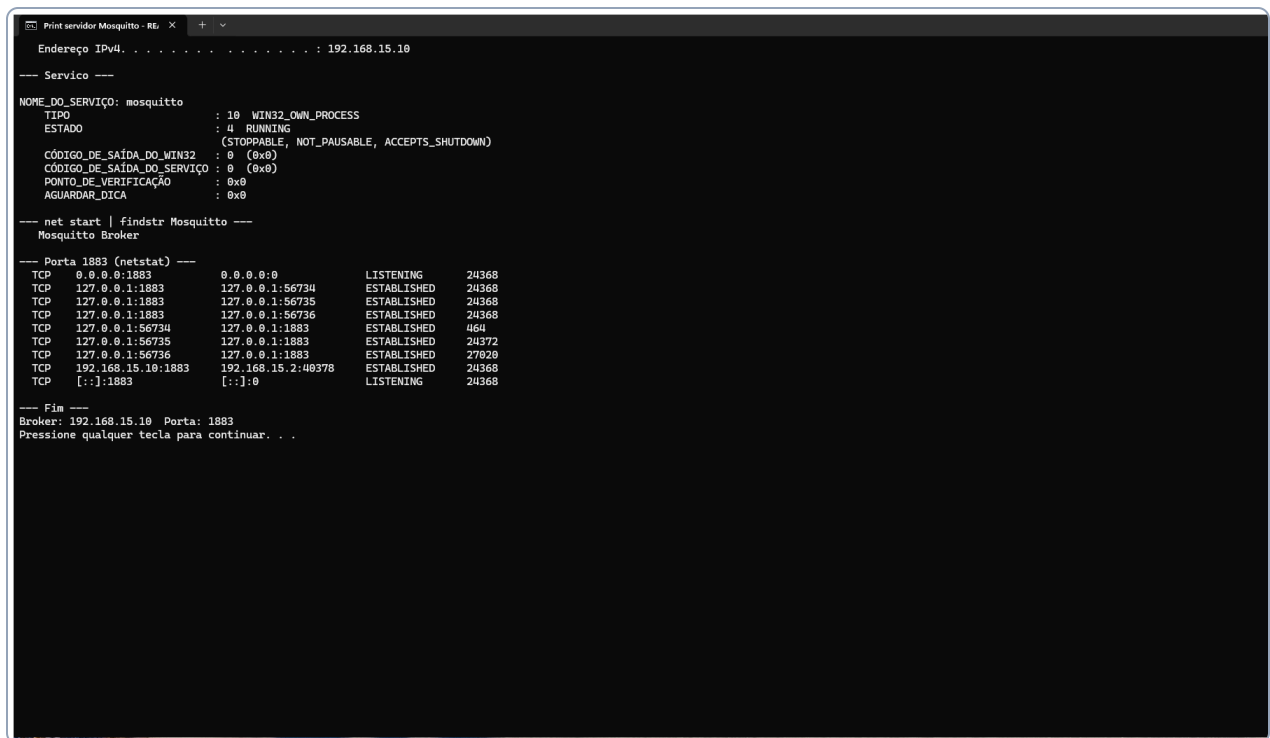


5.1 — Teste MQTT: celular ↔ Mosquitto

Broker: 192.168.15.10:1883

5.1 PC→celular: PASS · 5.2 celular→PC: PASS

1. Print REAL do servidor Mosquitto (Prompt)



```
Print servidor Mosquitto - RB X +
Endereço IPv4. . . . . : 192.168.15.10

--- Serviço ---
NOME_DO_SERVIÇO: mosquitto
TIPO           : 10  WIN32_OWN_PROCESS
ESTADO         : 4   RUNNING
                (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
CÓDIGO_DE_SAÍDA_DO_WIN32 : 0   (0x0)
CÓDIGO_DE_SAÍDA_DO_SERVIÇO : 0   (0x0)
PONTO_DE_VERIFICAÇÃO    : 0x0
AGUARDAR_DICA          : 0x0

--- net start | findstr Mosquitto ---
Mosquitto Broker

--- Porta 1883 (netstat) ---
TCP 0.0.0.0:1883 0.0.0.0:0 LISTENING 24368
TCP 127.0.0.1:1883 127.0.0.1:56734 ESTABLISHED 24368
TCP 127.0.0.1:1883 127.0.0.1:56735 ESTABLISHED 24368
TCP 127.0.0.1:1883 127.0.0.1:56736 ESTABLISHED 24368
TCP 127.0.0.1:56734 127.0.0.1:1883 ESTABLISHED 464
TCP 127.0.0.1:56735 127.0.0.1:1883 ESTABLISHED 24372
TCP 127.0.0.1:56736 127.0.0.1:1883 ESTABLISHED 27020
TCP 192.168.15.10:1883 192.168.15.2:40378 ESTABLISHED 24368
TCP [::]:1883 [::]:0 LISTENING 24368

--- Fim ---
Broker: 192.168.15.10 Porta: 1883
Pressione qualquer tecla para continuar. . .
```

Figura: Prompt real — sc query mosquitto RUNNING, IP 192.168.15.10, porta 1883 LISTENING + conexao do celular ESTABLISHED

2. Firewall / netstat / restart

```
netsh advfirewall firewall add rule name="Mosquitto MQTT" dir=in
action=allow protocol=TCP localport=1883
netstat -ano | findstr :1883
net stop mosquitto
net start mosquitto
```

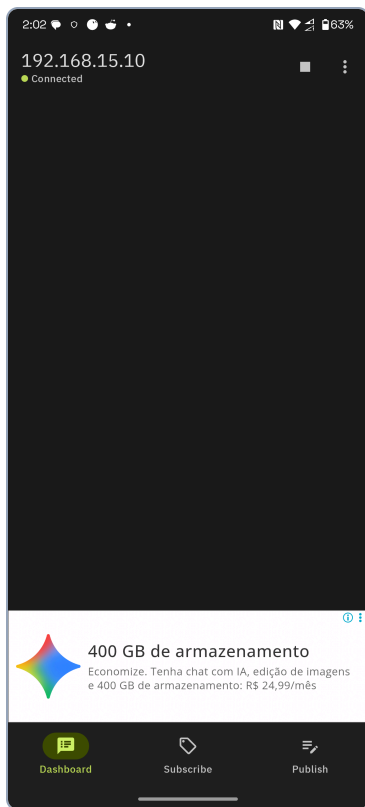


Figura: Print real — firewall Ok, netstat e reinício do serviço

3. Celular cliente — subscribe + mensagem do PC (5.1)

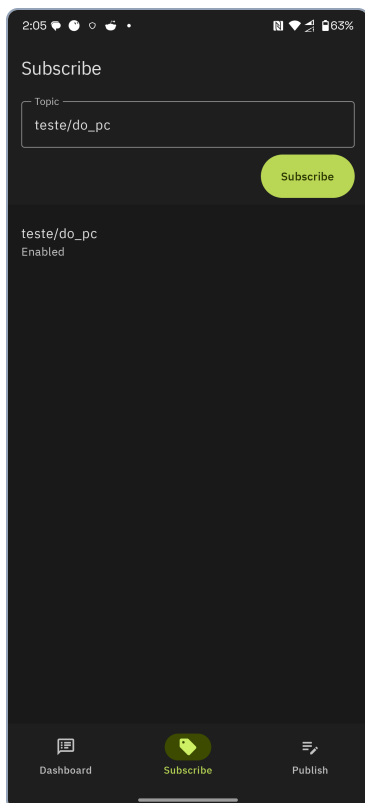


Figura: Celular — Subscribe Enabled em teste/do_pc

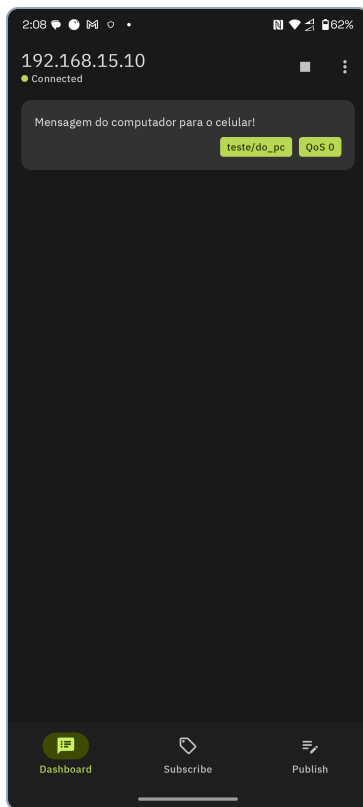


Figura: Celular Connected — mensagem recebida do PC (5.1 PASS)

4. Celular publica (5.2)

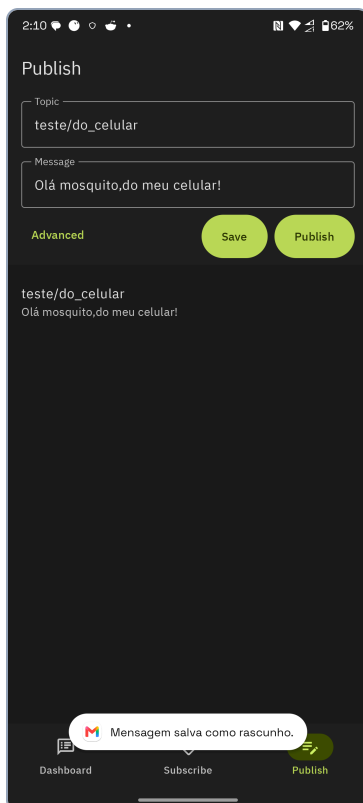


Figura: Celular — Publish em teste/do_celular (recebido no PC)

Checklist

Item	Status
Print real do servidor Mosquitto	OK
Print do celular como cliente	OK
Comandos firewall/netstat/restart	OK
Testes 5.1 e 5.2	PASS